NLP Pipeline Questions

Data Acquisition

- What do you understand by Data Acquisition?
- Why do we need to collect data?
- From where can we collect data for an NLP project?
- What makes a dataset good for solving a problem?

Text Preparation

- What do you understand by Text Preparation?
- Why is text cleaning important?
- What cleaning steps would you perform on raw text data?
- What is text preprocessing?
- What preprocessing techniques have you learned so far?
- When do we need advanced preprocessing techniques?

Regular Expressions (Regex)

- What is a Regular Expression (Regex)?
- Why do we use Regex in NLP?
- What is a regex pattern?
- What is the purpose of symbols like `.`, `*`, `+`, `?`, `^`, and `$`?
- How can Regex help us find email addresses, phone numbers, or URLs?
- How can Regex help in text cleaning?

Feature Engineering

- What do you understand by Feature Engineering?
- Why can't machine learning models work directly with raw text?
- When would you choose one text representation technique over another?

Modelling

- What do you understand by Modelling?
- How do you choose a machine learning algorithm for a problem?
- What algorithm would you use for a text classification task and why?
- What is the difference between training and testing a model?

Evaluation

Intrinsic Evaluation

- What do you understand by model evaluation?
- Why do we evaluate a model?

- What is Accuracy?
- What is Precision?
- What is Recall?
- What is F1-Score?
- When is Accuracy not a good metric?

Extrinsic Evaluation

- What do you understand by Extrinsic Evaluation?
- How can we determine whether a model is useful in a real-world application?
- Can a model have good accuracy but still fail in practice? Why?

Deployment

- What do you understand by Model Deployment?
- Why do we deploy a model?
- How can users interact with a deployed model?
- What challenges might occur after deployment?

Monitoring

- What do you understand by Model Monitoring?
- Why is monitoring important after deployment?
- What problems can occur if we do not monitor a model?
- What would you monitor in a deployed NLP system?

Model Update

- Why do models need updates?
- What is model drift?
- When should a model be retrained?
- How would you decide whether a new model is better than the current one?